Stick-Breaking Distributions and Pitman-Yor theorem

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Discrete Random Structures II

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Outline

Random discrete distribution

Standard Pitman-Yor process

Pitman-Yor theorem

Some remarks

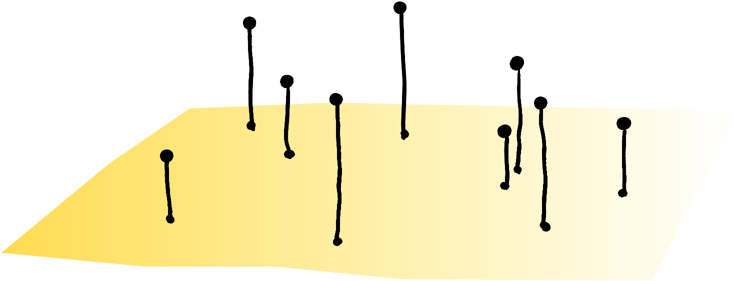
Random discrete distribution

$$\sum_i V_i \delta_{Q_i}$$

where

$$\sum_i V_i = 1, \quad V_i \geqslant 0\,.$$

(V_i) is called a (random) **mass partition**.

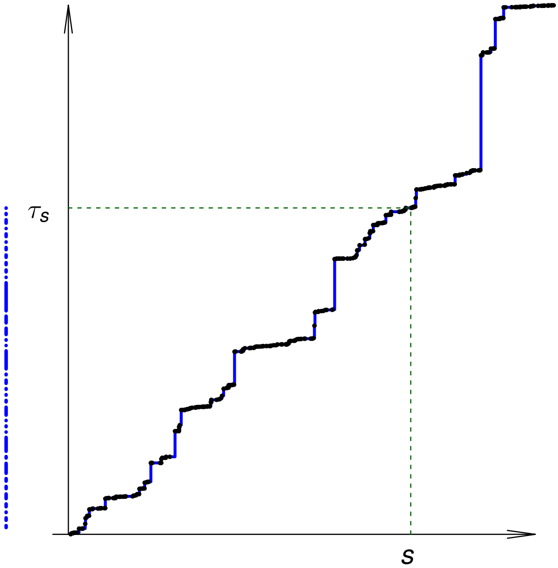


Applications

- excursion theory (Perman, Pitman, Yor),
- genetics (Crane),
- combinatorial stochastics (Arratia and Tavare, Pitman, Gnedin),
- statistical physics (Derrida),
- Bayesian nonparametrics and machine learning (Ayed, De Blasi et al...)

How do you generate a mass partition?

Subordinators



Let N be a Poisson point processes (PPP) with intensity measure $Leb \times \mu$

$$N = \sum_{i=1}^{\infty} \delta_{T_i,P_i} \sim \mathrm{PPP}(Leb \times \mu)\,.$$

with $\int (x \wedge 1)\mu(dx) < \infty$, then

$$\tau_t = \sum_{T_i \leqslant t} P_i$$

is a subordinator with no drift and Lévy measure μ.

Self-normalization map

Works on summable point processes on $[0, \infty)$ as

$$\mu = \sum_i \delta_{x_i} \mapsto \mathcal{V}(\mu) = \sum_i \delta_{x_i / \sum_j x_j}.$$

or on summable sequences in $[0, \infty)$

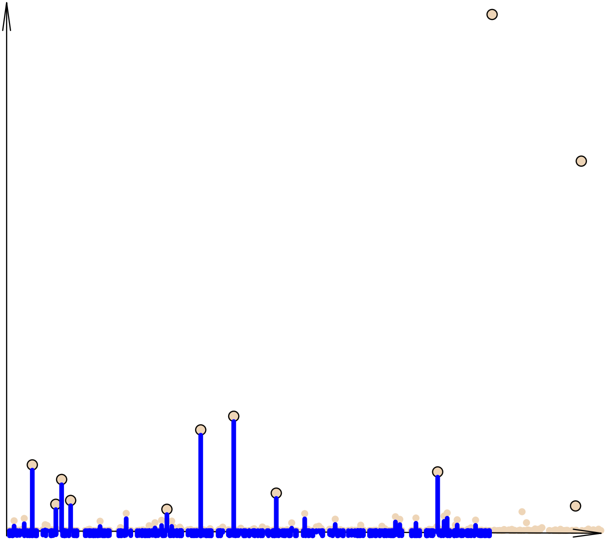
$$(x_n) \mapsto \mathcal{V}(x_n) = \left(\frac{x_n}{\sum_i x_i} \right) ,$$

Clearly self-normalized jumps of a subordinator until time $s > 0$ generate random distribution.

Main choices (Kingman, 1975) for μ corresponding to the (standard) Poisson-Dirichlet and Pitman-Yor distributions are

- $\mu(dt) = \frac{e^{-t}}{t} dt$
- $\mu(dt) = \mu_\alpha(dt) = \frac{K}{t^{\alpha+1}} dt$ for a fixed $K > 0$ and $\alpha \in (0, 1)$

We abbreviate this as sPD and sPY, or alternatively as PD$(0, s)$ and PD$(\alpha, 0)$ distributions.



$$\sum_{T_i \leqslant s} \delta_{P_i}$$

Stick-breaking distributions

Assume (Y_i) take values in $[0, 1]$. Consider now

$$V_1 = Y_1, \; V_2 = (1 - Y_1)Y_2, \;\; V_3 = (1 - Y_1)(1 - Y_2)Y_3,$$
$$\ldots, V_n = (1 - Y_1)(1 - Y_2)\cdots(1 - Y_{n-1})Y_n, \ldots$$

The sequence (V_n) sums up to one under very mild conditions, e.g. if (Y_j) is a nondegenerate i.i.d. sequence.

They are also called general **residual allocation models** (RAM), *beta*-stick-breaking was studied by Ishwaran and James (2001).

Perpetuities

If $G_i \sim gamma(\alpha_i, 1)$, $i = 1, \ldots, n$, are independent then r. vector $\mathcal{V}(G_1, \ldots, G_n)$ has **Dirichlet distribution**

For $n = \infty$ it was suggested to consider i.i.d. (say gamma) G_i's and

$$\mathcal{V}\left(G_n \Pi_n\right)$$

for some independent nonnegative weights (Π_n).

Consider therefore

$$\mathcal{V}(G_n \Pi_n) \quad \text{with} \quad \Pi_n = \prod_{i=1}^{n-1} U_i,\ j \geqslant 1,$$

for a sequence (U_n) of independent nonnegative random variables independent of (G_n).

If U_n's are i.i.d. and if $\mathbb{E}\log U < 0$ and if $\mathbb{E}\log^+ G < \infty$, then

$$R = \sum_n G_n \Pi_n < \infty$$

and

$$R \overset{d}{=} UR + G\,,$$

see Vervaat (1979) or Buraczewski, Damek, Mikosch (2016).

beta/gamma (perpetuity) family

Suppose

$$\sum_i \delta_{V_i} \stackrel{d}{=} \mathcal{V}(G_n \Pi_n)$$

where $\Pi_n = \prod_{i=1}^{n-1} U_i$, for some independent random variables

$$U_i \sim \textit{beta} \quad \text{and} \quad \text{iid} \;\; G_i \sim \textit{gamma}$$

- Standard PD and PY distributions both belong to this class.

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Recall $N \sim \mathrm{PPP}(Leb \times \mu)$ where we choose $K = \alpha/\Gamma(1-\alpha)$ in

$$\mu(dt) = \mu_\alpha(dt) = \frac{K}{t^{\alpha+1}} dt$$

Mark points of the process $N = \sum_i \delta_{T_i, P_i}$ with iid exponentials to arrive at

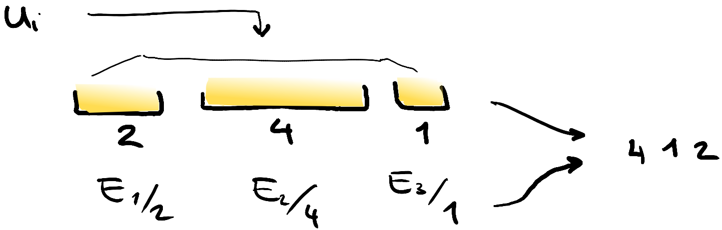
$$N^E = \sum_i \delta_{T_i, P_i, E_i} \, .$$

And transform it to another PPP

$$N^{E/P} = N_s^{E/P} = \sum_{T_i \leqslant s} \delta_{E_i/P_i, P_i} \, .$$

Fix $s = 1$ in the sequel.

Size-biasing



Size-biasing

Comparing intensity measures we get

$$N^{E/P} = \sum_{T_i \leqslant s} \delta_{E_i/P_i, P_i} = \sum_{j \geqslant 1} \delta_{Z_j, G_j/Z_j} = \sum_{j \geqslant 1} \delta_{Z_j, \tilde{P}_j}$$

where $(Z_j)_{j \geqslant 1} \stackrel{d}{=} (\Gamma_j^{1/\alpha})_{j \geqslant 1}$ for standard PPP and (G_j) is an independent i.i.d. gamma$(1 - \alpha, 1)$ sequence. Obviously

$$\sum \delta_{P_i} = \sum \delta_{\tilde{P}_j}$$

Turns out $(\tilde{P}_j)$ is a (random) size-biased ordering of $\{P_i\}$.

Thus $\{V_i\}$ have size-biased representation

$$(\tilde{V}_n) = \mathcal{V}\left(\tilde{P}_n\right) = \mathcal{V}(G_n/Z_n) = \mathcal{V}(G_n\Pi_n),$$

where $G_i \sim gamma(1-\alpha, 1)$ and $\Pi_j = \prod_{i=1}^{j-1} Z_i/Z_{i+1}$.

Note,

$$U_i = Z_i/Z_{i+1} = (\Gamma_i/\Gamma_{i+1})^{1/\alpha} \sim beta(i\alpha, 1)$$

are independent random variables.

Model

$$\text{For} \quad \alpha \geqslant 0,\ s > 0 \quad \text{and} \quad c > 0,$$

$$\text{let} \quad a_i = s + (i-1)\alpha,\ i \geqslant 1\,.$$

Take independent

$$U_i \sim beta(a_i, c+\alpha) \quad \text{and} \quad G_i \sim gamma(c, 1)\,.$$

For $\Pi_j = \prod_{l=1}^{j-1} U_l$, calculate

$$\mathcal{V}(G_n \Pi_n)$$

Standard distributions

$$sPY: \quad \alpha = s \in (0,1) \quad \text{and} \quad c = 1-\alpha$$
$$sPD: \quad \alpha = 0,\ s > 0, \quad \text{and} \quad c = 1\,.$$

Theorem

Under this model, series $\sum_j G_j \Pi_j$ *converges a.s. and*

$$(\tilde{V}_n) = \mathcal{V}(G_n \Pi_n)$$

has the representation

$$\tilde{V}_1 = \tilde{Y}_1, \quad \tilde{V}_n = (1 - \tilde{Y}_1) \cdots (1 - \tilde{Y}_{n-1}) \tilde{Y}_n, \quad n \geqslant 2\,,$$

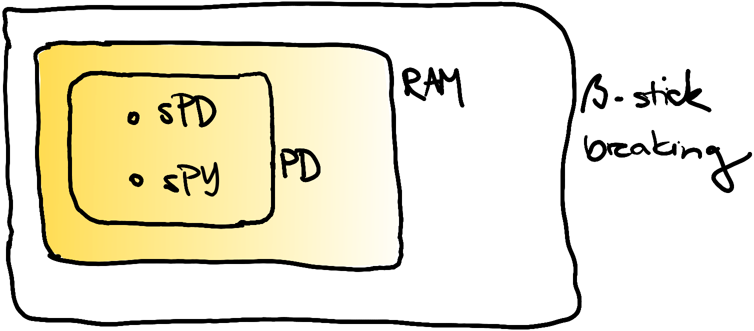
with independent $\tilde{Y}_n \sim \textit{beta}(c, a_n)$.

Under this model, we say the distribution of the random mass partition

$$(\tilde{V}_n) = \mathcal{V}(G_n \Pi_n)$$

has a beta/gamma perpetuity representation with parameters (α, s, c). Therefore it also has residual allocation representation — we denote this by $\mathrm{RAM}(\alpha, s, c)$.

One can show all other PD(α, θ) distributions belong to this family, so they all have stick breaking representation (McCloskey (1965), Perman (1992), ...)


sPD
sPY
PD
RAM
B-stick
breaking

All other $PD(\alpha, s)$ are obtained by change of measure from sPY. Our analysis shows that they satisfy

$$(\tilde{V}_n) = \mathcal{V}\left(\frac{G_n}{(D_1' + D_2 + \cdots + D_n)^{1/\alpha}}\right),$$

where $D_1' \sim gamma(s/\alpha + 1, 1)$ is independent of the i.i.d. standard exponential (D_i).

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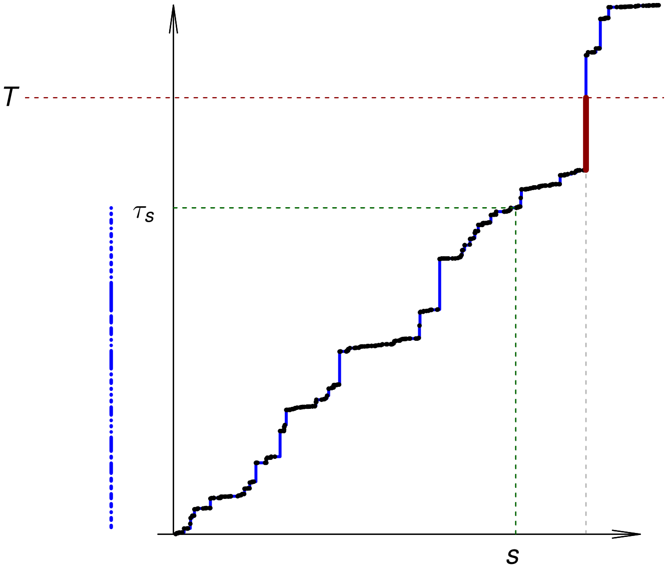
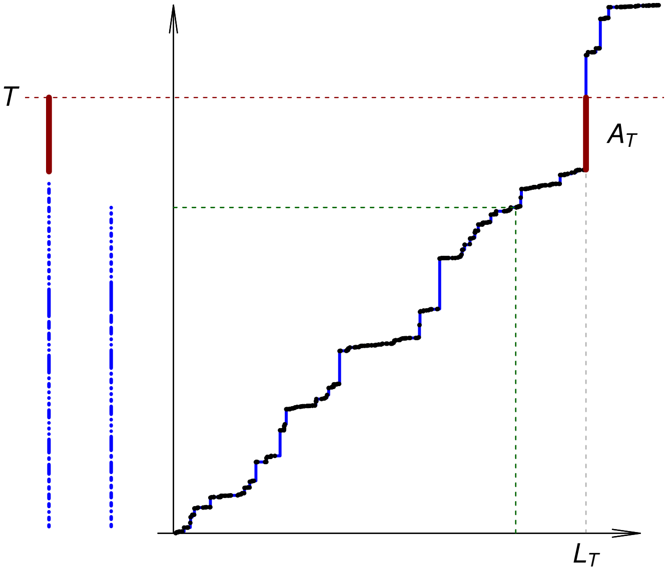
Let N be a $\mathrm{PPP}(Leb \times \mu_\alpha)$. We studied mass partition corresponding to the jumps of stable subordinator over time, i.e.

$$\eta = \eta(s) = \mathcal{V}\left(\sum_{T_i \leqslant s} \delta_{P_i}\right) = \mathcal{V}(G_n/Z_n).$$

One can alternatively consider closure of the range of the subordinator intersected with segment $[0, T]$.



T
τ_s
s



T
A_T
L_T

Inverse of the α-stable subordinator $\tau_t = \sum_{T_i \leqslant t} P_i$, $t \geqslant 0$, is

$$L_T = \inf\{t : \tau_t > T\}\,,$$

The overshoot of the jump over T is $B_T = \tau(L_T) - T$, the under-shoot is

$$A_T = T - \tau(L_T -).$$

Denote

$$\xi = \xi(T) = \{P_i/T : \; T_i < L_T\} \cup \{A_T/T\}.$$

Mysteriously ξ and η are equal in distribution.

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Mysteriously ξ and η are equal in distribution.

Theorem (Pitman and Yor)

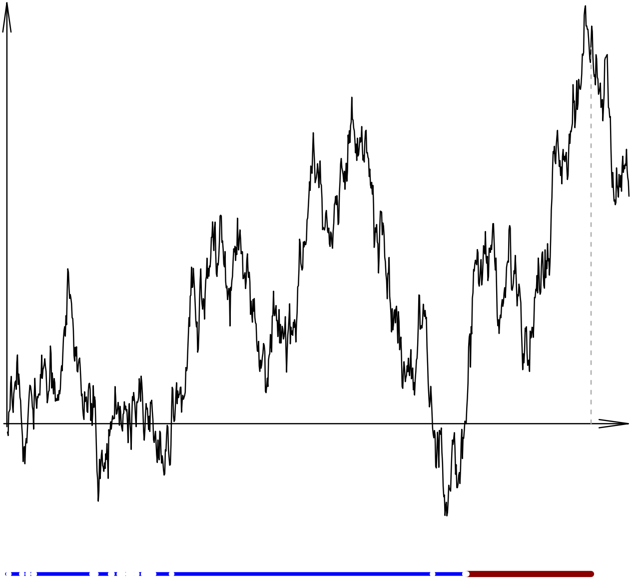
For any fixed $\alpha \in (0, 1)$*, and all constants* $K, \ T, \ s > 0$

$$\xi(T) \stackrel{d}{=} \eta(s)\,.$$

Bessel process

Lévy showed that for $\alpha = 1/2$ jumps of (τ_t) correspond to the excursions of Brownian motion.

Connection between the excursions of Bessel processes and (τ_t) (and the process N therefore) was established by Molchanov and Ostrovskii (1969).



Thinnings

Recall we marked N to obtain

$$N^E = \sum_i \delta_{T_i, P_i, E_i} \,.$$

One can clearly write $N = N^\wedge + N^\vee$ for independent $\mathrm{PPP}s$

$$N^\wedge = \sum_i \delta_{T_i^\wedge, P_i^\wedge} = \sum_i \delta_{T_i, P_i} \mathbb{1}_{P_i > E_i}$$

$$N^\vee = \sum_i \delta_{T_i^\vee, P_i^\vee} = \sum_i \delta_{T_i, P_i} \mathbb{1}_{P_i \leqslant E_i} .$$

Furthermore, in

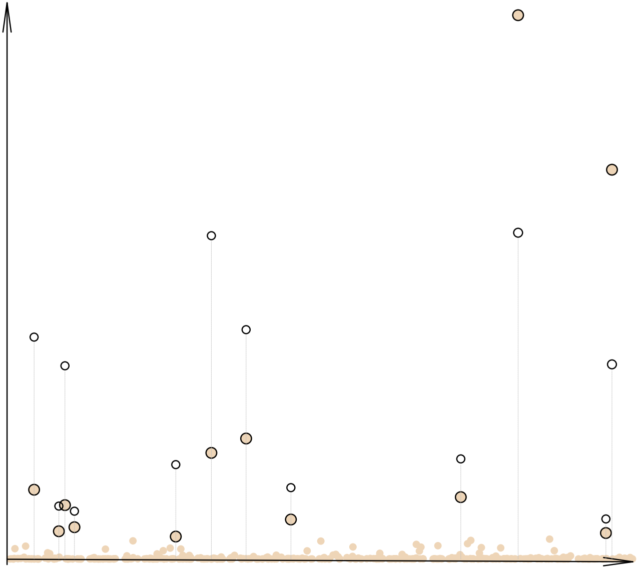
$$N^{\wedge,E} = \sum_i \delta_{T_i,P_i,E_i} \mathbb{1}_{P_i > E_i} = \sum_n \delta_{T_n^\wedge,P_n^\wedge,E_n^\wedge}$$

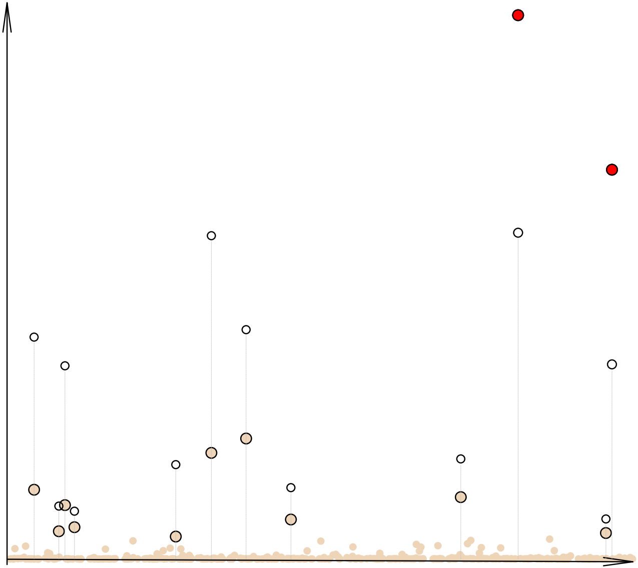
$$(T_n^\wedge) \stackrel{d}{=} (\Gamma_n)$$

is independent of i.i.d. pairs

$$(P_n^\wedge, E_n^\wedge)_n \stackrel{d}{=} (H_n E_n^\wedge, E_n^\wedge)_n$$

where $E_n^\wedge \sim \mathit{gamma}(1-\alpha, 1)$ are independent of $H_n \sim \mathit{Pareto}(\alpha)$ (cf. Bertoin et al 2006 – BFRY distribution, Devroye and James 2014)





Proof.

N satisfies the scaling property for any fixed $c > 0$

$$N \stackrel{d}{=} \sum_i \delta_{cT_i, c^{1/\alpha} P_i} \,.$$

Hence, distributions of $\eta(s)$ and $\xi(T)$ are independent of s, T and the constant K in

$$\mu_\alpha(dt) = \frac{K}{t^{\alpha+1}} dt.$$

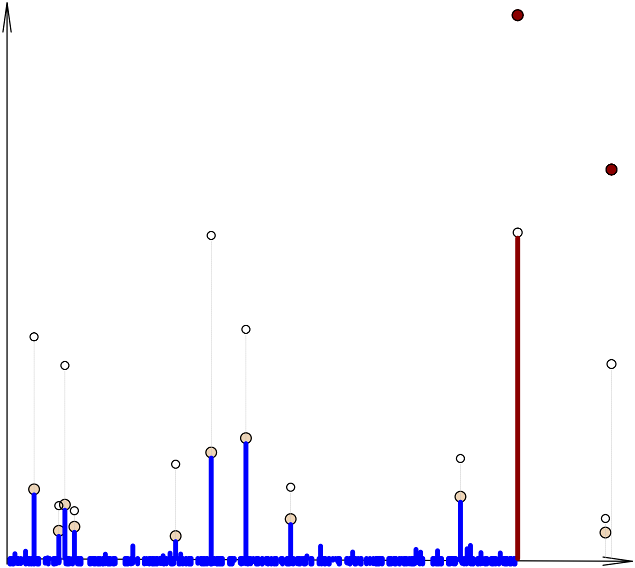
Denote

$$X = \sum_{T_i^{\vee} \leqslant T_1^{\wedge}} P_i^{\vee} + E_1^{\wedge} .$$

By the definition $T_1^{\wedge} = L_X$ a.s. Moreover X is standard exponential and independent of N.

Thus, it suffices to show

$$\xi(X) \stackrel{d}{=} \eta(1).$$



Observe that

$$A_X = E_1^\wedge \stackrel{d}{=} G_1 \sim gamma(1-\alpha, 1) \tag{1}$$

is independent of $T_1^\wedge \stackrel{d}{=} \Gamma_1$ and $N^\vee$.

Turns out that

$$N^{\vee}_{T_1^{\wedge}} \stackrel{d}{=} \left\{ G_n \left(\frac{\Gamma_1}{\Gamma_n} \right)^{1/\alpha} \right\}_{n \geqslant 2} .$$

Thus,

$$\left(N^{\vee}_{T_1^{\wedge}} ; A_X \right) \stackrel{d}{=} \left(\left\{ G_n \left(\frac{\Gamma_1}{\Gamma_n} \right)^{1/\alpha} \right\}_{n \geqslant 2} ; G_1 \right)$$
$$= \left(\left\{ G_n \frac{Z_1}{Z_n} \right\}_{n \geqslant 2} ; G_1 \right)$$

with two components independent of each other on both sides.

It remains to observe that

$$\xi(X) = \mathcal{V}\left(N^{\vee}_{T_1^{\wedge}} \cup \{A_X\}\right)$$

$$\eta(1) = \mathcal{V}\left(\left\{G_n \frac{Z_1}{Z_n}\right\}_{n \geqslant 2} \cup \{G_1\}\right) = \mathcal{V}(G_n/Z_n).$$

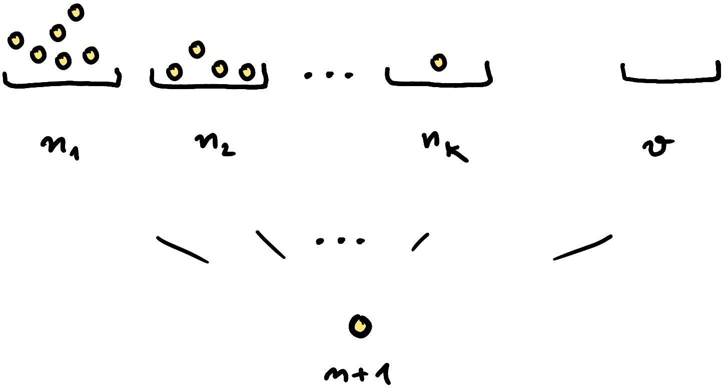
$\square$

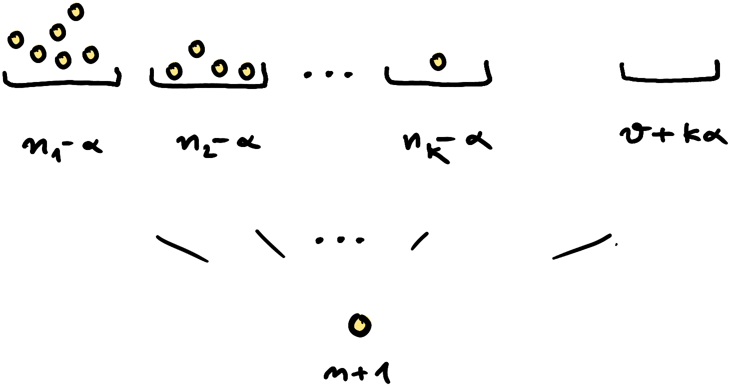
Outline

Remarks

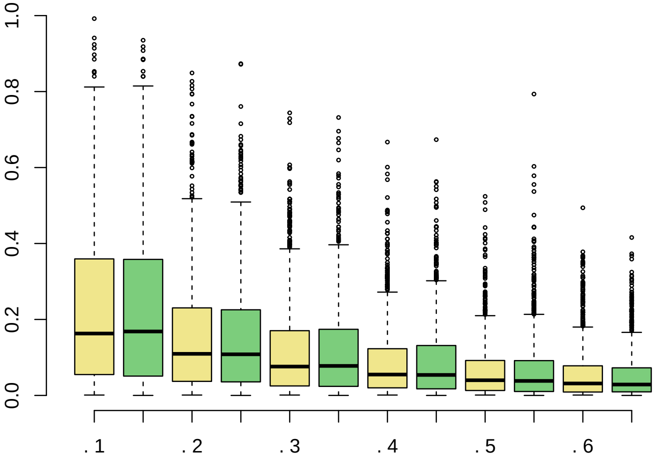
- all *PD* distributions belong to beta/gamma perpetuity mass partitions which all have stick breaking representation
- this provides an alternative representation and extension of *PD*$(\cdot, \cdot)$ class covering any $\alpha \geqslant 0$
- a new thinning proof of Pitman-Yor theorem
- suggests a new version of Chinese Restaurant Process
- an updated formulation of arcsine laws for Brownian motion and Bessel processes

CRP / Hoppe urn





$n_1 - \alpha$
$n_2 - \alpha$
$\cdots$
$n_k - \alpha$
$\theta + k\alpha$
$n+1$



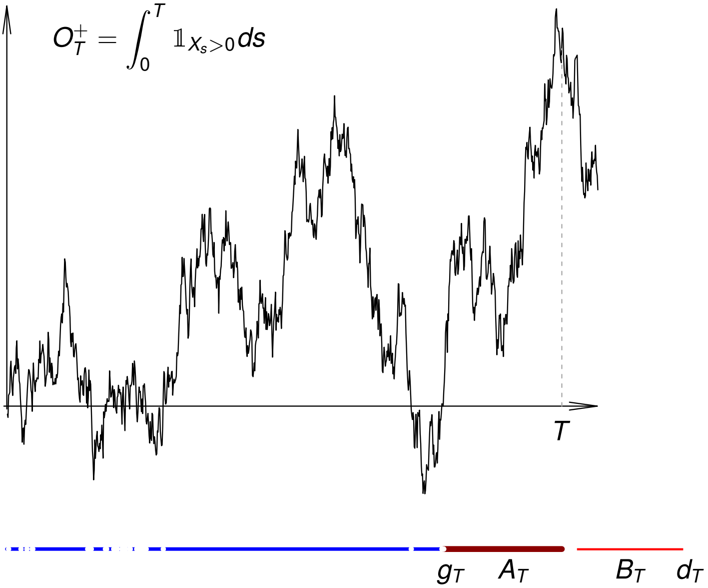
500 simulations: relative occupancy of tables vs $\tilde{V}_1, \ldots, \tilde{V}_6$

Arcsine laws

Suppose (X_s) denotes a standard Brownian motion, let

$$g_T = \sup\{s < T : X_s = 0\} \quad \text{and} \quad d_T = \inf\{s > T : X_s = 0\}$$

$$\Delta_T = A_T + B_T \quad \text{and} \quad O_T^+ = \int_0^T \mathbb{1}_{X_s > 0} ds \,.$$


$O_T^+ = \int_0^T \mathbb{1}_{X_s>0} ds$
T
g_T
A_T
B_T
d_T

Corollary

For any $T > 0$

$$\left(\frac{O_T^+}{T}, \frac{A_T}{T}, \frac{B_T}{T}, \frac{g_T}{T}, \frac{d_T}{T}, \frac{\Delta_T}{T}, \sum_{T_i < L_T} \delta_{P_i/T}\right) \stackrel{d}{=} \big(\varepsilon(Q,\eta'), 1-Q, (1-Q)(H-1), Q, Q+(1-Q)H, (1-Q)H, Q\eta'\big)\,,$$

where $Q \sim$ *beta*$(1/2, 1/2)$, $H \sim$ *Pareto*$(1/2)$, $\eta' \sim$*PD*$(1/2, 1/2)$ *and an i.i.d. Bernoulli sequence* (ε_n) *on the right hand side are independent.*

Here $\eta' = \frac{1}{1-\tilde{V}_1}\{\tilde{V}_n\}_{n\geqslant 2}$ and $\varepsilon(Q,\eta') = (1-Q)\varepsilon_1 + \sum_{i\geqslant 2} \tilde{V}_i \varepsilon_i$.